

Aman Kumar

Backend Developer

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🌐 LinkedIn

📁 Portfolio

🐙 Github

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Summary

Backend Developer with 3+ years of experience building scalable backend systems, REST APIs, and data-intensive applications. Strong in Python, Django, PostgreSQL, Redis, AWS, SQL optimization, and backend architecture, with hands-on experience delivering cloud-native systems handling large enterprise datasets and high-volume workloads.

Professional Experience

Genpact

Aug 2023 – Present

Software Developer

Bengaluru, India

- Designed and scaled Python and Django backend services processing **500,000+ requests per month**, improving throughput and supporting low-latency enterprise workloads.
- Optimized complex SQL queries, indexing strategies, and data access patterns for large-scale datasets, reducing execution time by **60%**.
- Built a Redis-based caching layer using the Cache-Aside pattern and LRU eviction, reducing database read load by **40%** on high-traffic endpoints.
- Client Project: GE Vernova (Data Archival Portal)** — *Python, Django REST API, AWS Lambda, S3, Athena*
 - Architected a low-cost archival platform that migrated metadata to **AWS S3** cold storage, reducing storage costs by **80%** across **230** enterprise projects while improving long-term scalability and retention management.
 - Designed secure backend access workflows for **US Export Controlled (US EC)** data using **Python, Django REST APIs, AWS Lambda, Athena, and Row-Level Security (RLS)**, enabling compliant retrieval and governed querying across **6M+ files and records** in a shared multi-tenant environment.
 - Implemented **trigram-based filtering/search** for encrypted Athena tables and integrated **AWS OpenSearch** for full-text discovery across **3M+ encrypted files**, building paginated, high-performance APIs to improve secure data discoverability and retrieval at scale.
 - Designed and integrated **AWS OpenSearch (Elasticsearch)** for full-text search across **3M+ encrypted files**, and built paginated **Django REST APIs** for scalable, low-latency, and reliable data retrieval.
- Client Project: GE Vernova (Data Access Portal)** — *Python, Django REST API, SQL, Cloud Datasets*
 - Developed a financial visualization platform with a natural language interface used by **100+ users** to generate custom JSON templates, improving analysis speed by **30%**.
 - Built SQL-to-JSON query translation and validation pipelines over cloud datasets, improving query accuracy and reducing manual validation effort.
 - Automated migration scripts for transition from EY to GE Vernova, importing **130+ blank templates** and **10,000+ reports** with minimal downtime.
- Client Project: GE Vernova (Vernova Data Portal)** — *Python, Django REST API, SQL, AWS*
 - Revamped the backend of a legacy data portal by implementing a hierarchical domain-based access model and integrating chatbot-enabled reporting workflows with embedded BI dashboards.

Enquero Global LLP (A Genpact Company) [Certificate]

Mar 2023 – Jul 2023

Software Engineering Intern

Bengaluru, India

- Built RESTful APIs for subscription management, session booking, and vitals monitoring using Python and Django, contributing to a **25%** increase in user engagement.

Skills

Languages: Python, SQL, C++

Backend: Django, Django REST Framework, Node, REST APIs, Microservices

Databases & Caching: PostgreSQL, MySQL, Redis, AWS Athena, SQL Optimization, NoSQL

Cloud & DevOps: AWS, Docker, Kubernetes, OpenSearch, Linux, Nginx,

Background Processing: Asynchronous Workflows, Message Queues, Celery, RabbitMQ, Task Scheduling

Core CS & Architecture: System Design, Distributed Systems, Data Structures & Algorithms

Certificates and Achievements

Microsoft Certified: Azure Fundamentals (AZ-900) [Certificate]

Databricks Certified DE Associate [Certificate]

Agentic AI - Fundamentals [Certificate]

Generative AI - Fundamentals [Certificate]

HackWithInfy 2022 Qualified

Coding Profiles: [LeetCode](#) — [GeeksForGeeks](#)

Education

Visvesvaraya Technological University (VTU)

2019 – 2023

Bachelor of Engineering in Computer Science — CGPA: 8.51/10.0

Bengaluru, India